DS-A4 Model Comparison and Calibration

Data and frozen protocol

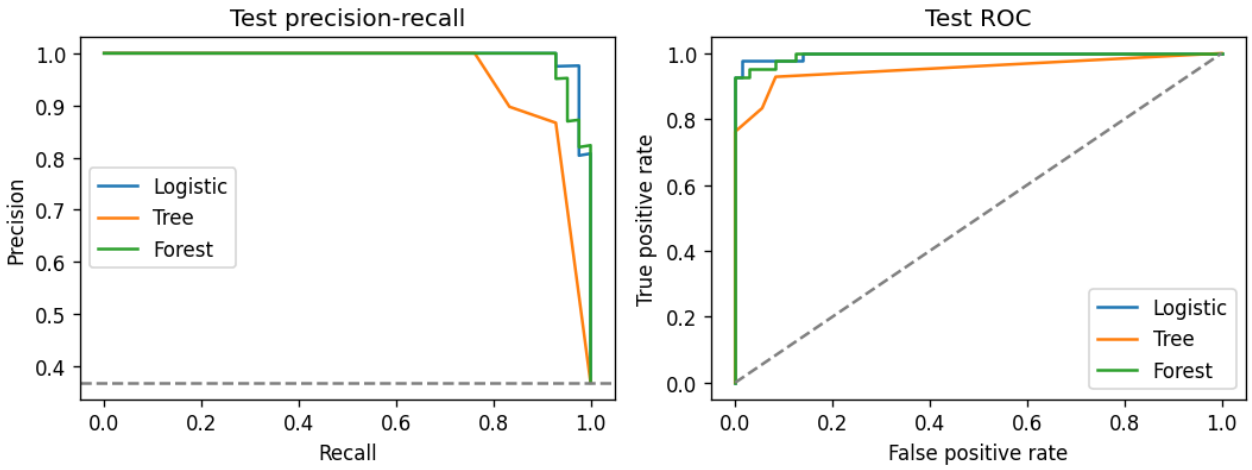
The sklearn Breast Cancer Wisconsin dataset has 569 samples, 30 numeric features and no missing values. Malignant is coded as 1. Its small size and complete inputs suit a reproducible model comparison.

Use an 80/20 stratified split (455 training, 114 test), five identical training folds and seed 42. Each model gets one configuration: Logistic C=1 with fold-local scaling; Tree depth 3/leaf 10; Forest 200 trees/depth 5/leaf 5. Select by pooled training OOF AP before test evaluation.

Model comparison

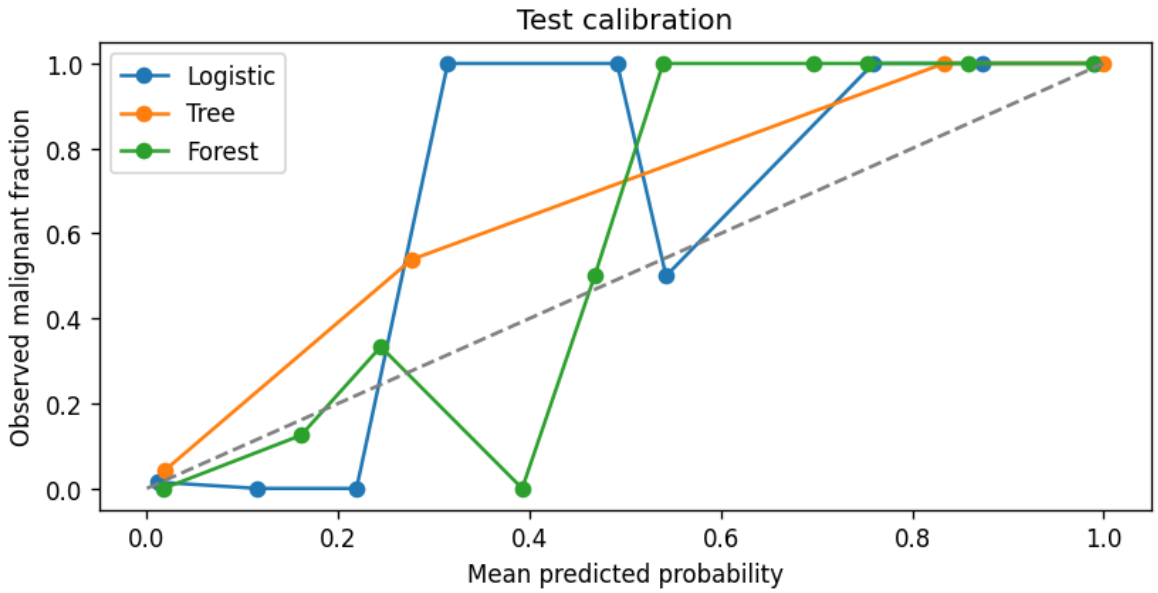
Model	OOF AP	Test AP	ROC AUC	Brier	ECE
Logistic	0.9943	0.9943	0.9960	0.0213	0.0302
Tree	0.9313	0.9349	0.9527	0.0632	0.0533
Forest	0.9849	0.9916	0.9944	0.0341	0.0516

Logistic is selected. The complex models do not improve AP or Brier. Higher AP/ROC AUC and lower Brier/ECE are better; AP summarizes precision-recall performance.



Full code, paired predictions and statistics are in main.ipynb and results.csv.

Calibration and uncertainty



Calibration uses ten equal-width bins. Sparse test bins are noisy; ECE depends on binning. Brier reflects discrimination and calibration. The 95% intervals use 1,000 paired, class-stratified test bootstraps and fixed fitted models.

Model	AP 95% interval	ROC 95% interval	Brier 95% interval	ECE 95% interval
Logistic	0.9821 to 1.0000	0.9868 to 1.0000	0.0066 to 0.0414	0.0203 to 0.0607
Tree	0.8740 to 0.9811	0.9019 to 0.9908	0.0313 to 0.0983	0.0231 to 0.0947
Forest	0.9771 to 1.0000	0.9838 to 1.0000	0.0168 to 0.0547	0.0438 to 0.0879

Paired differences in AP

Difference	Value	95% low	95% high
Tree - Logistic	-0.0594	-0.1141	-0.0164
Forest - Logistic	-0.0026	-0.0138	0.0062

Forest-minus-Logistic includes zero: this sample does not establish a ranking-performance difference.

Critical slices and counterexample

The training median radius is 13.34. The large-radius group has 58 rows, 40 positives, one FN and one FP; the small-radius group has 56 rows, only two positives, and no errors. Class mix limits this comparison.

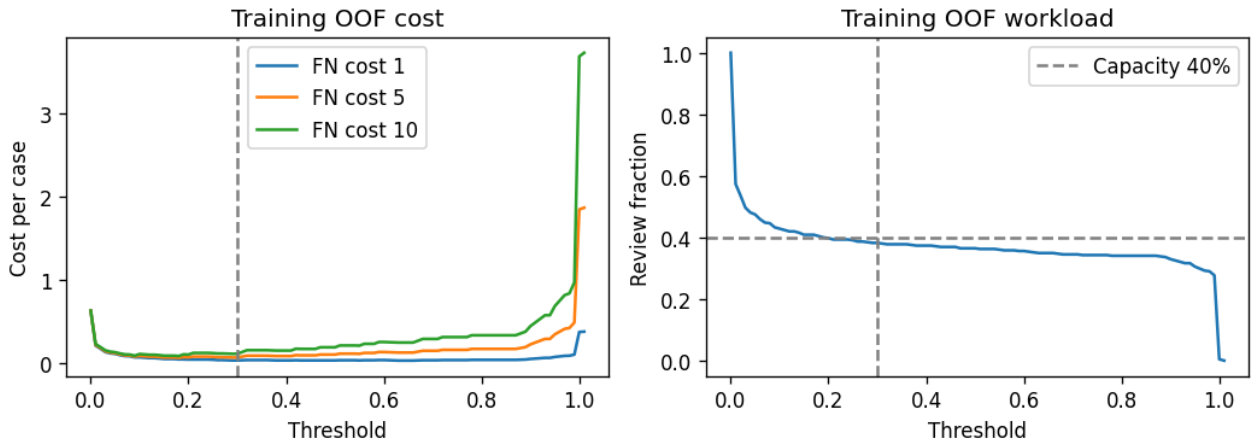
Row	Actual	Probability	Decision	Failure
73	1	0.0614	0	FN
455	0	0.5092	1	FP

Row 73 is malignant but receives probability 0.0614 and is missed. Strong average metrics do not guarantee every individual decision.

Threshold and sensitivity

Assume false-negative cost 5, false-positive cost 1 and review capacity 40%. On training OOF predictions, minimize (5 FN + FP)/n subject to the review-rate limit. Ties favor fewer reviews, then a higher threshold. The frozen grid is 0 to 1.01 in steps of 0.01.

Choose threshold 0.30: OOF review rate 38.24% and cost 0.0615 per case. On the test set it flags 42/114 cases (36.84%), with one FN and one FP, giving cost 6/114 = 0.0526 per case. Costs and capacity are illustrative.



FN cost	Cap	t	OOF review	OOF cost	Test review	FN	FP	Test cost
1	0.2000	1.0000	0.0044	0.3692	0.0000	42	0	0.3684
1	0.3000	0.9600	0.2989	0.0747	0.2632	12	0	0.1053
1	0.4000	0.6600	0.3495	0.0242	0.3333	4	0	0.0351
5	0.2000	1.0000	0.0044	1.8462	0.0000	42	0	1.8421
5	0.3000	0.9600	0.2989	0.3736	0.2632	12	0	0.5263
5	0.4000	0.3000	0.3824	0.0615	0.3684	1	1	0.0526
10	0.2000	1.0000	0.0044	3.6923	0.0000	42	0	3.6842
10	0.3000	0.9600	0.2989	0.7473	0.2632	12	0	1.0526
10	0.4000	0.2000	0.3978	0.0967	0.3947	1	4	0.1228

At 20% capacity, the 0.01 grid selects 1.00 and severely underuses capacity near probability 1. This poor setting is retained. Higher FN costs at 40% capacity lower the threshold and increase workload.

Conclusion and reproduction

Use Logistic at threshold 0.30 for the assumed 5:1 cost ratio and 40% expected capacity. The small test set, one cohort, sparse calibration bins and unequal subgroup class mixes limit generalization. Intervals do not include model-training uncertainty.

Python 3.12.14; sklearn 1.7.2. Install requirements.txt, restart the kernel and run main.ipynb. Raw OOF/test results are saved in results.csv; README includes commands and the submission summary. AI use is recorded in AI_USE.md.

Data: scikit-learn.org/stable/modules/generated/sklearn.datasets.load_breast_cancer.html